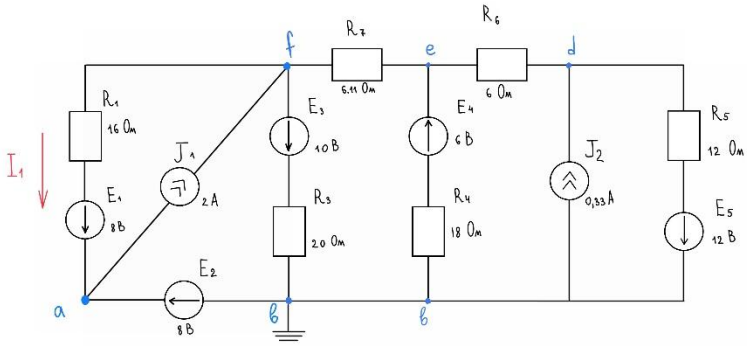
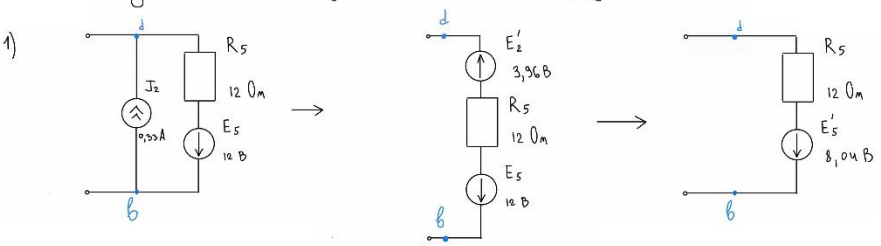
1. Рассчитать ток в одной из ветвей методом эквивалентных преобразований



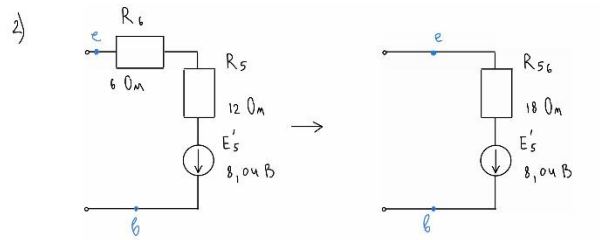

Найдем ток $I_1$ методом Эквивалентных преобразований:

1)



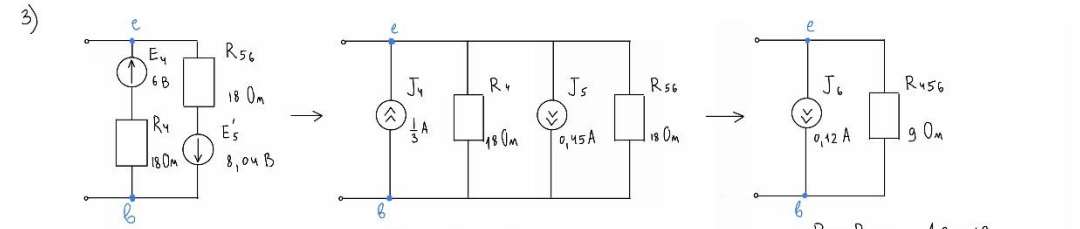

$E_2' = J_2 R_5 = 0{,}33 \cdot 12 = 3{,}96$ В $\qquad E_5' = E_5 - E_2' = 12 - 3{,}69 = 8{,}04$ В

2)




$R_{56} = R_5 + R_6 = 12 + 6 = 18$ Ом

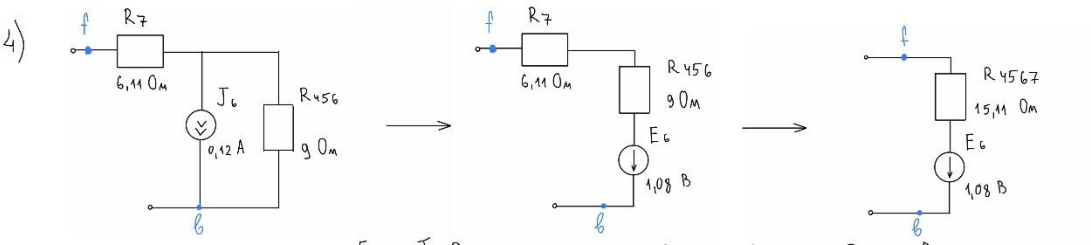
3)




$J_4 = \frac{E_4}{R_4} = \frac{6}{18} = \frac{1}{3}$ А $\qquad R_{456} = \frac{R_4 \cdot R_{56}}{R_4 + R_{56}} = \frac{18 \cdot 18}{18 + 18} = 9$ Ом

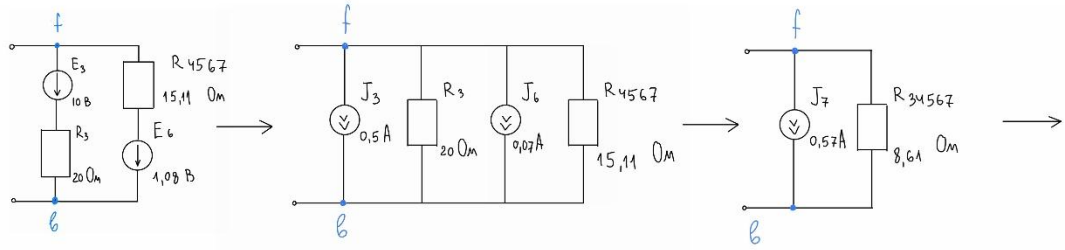
$J_5 = \frac{E_5'}{R_{56}} = \frac{8{,}04}{18} \approx 0{,}45$ А $\qquad J_6 = J_5 - J_4 = 0{,}45 - \frac{1}{3} \approx 0{,}12$ А

4)




$E_6 = J_6 R_{456} = 0{,}12 \cdot 9 = 1{,}08$ В $\qquad R_{4567} = R_{456} + R_7 = 9 + 6{,}11 = 15{,}11$ Ом

5)

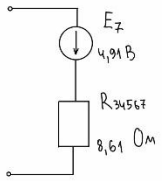



$$J_3 = \frac{E_3}{R_3} = \frac{10}{20} = 0{,}5 \text{ A}$$

$$J_6 = \frac{E_6}{R_{4567}} = \frac{1{,}08}{15{,}11} \approx 0{,}07 \text{A}$$

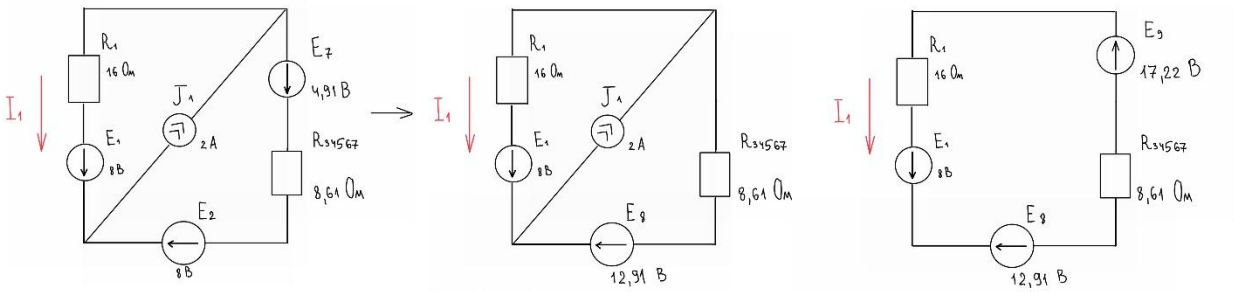
$$J_7 = J_3 + J_6 = 0{,}5 + 0{,}07 = 0{,}57 \text{ A}$$

$$R_{34567} = \frac{R_3 \cdot R_{4567}}{R_3 + R_{4567}} = \frac{20 \cdot 15{,}11}{20 + 15{,}11} \approx 8{,}61 \text{ Ом}$$



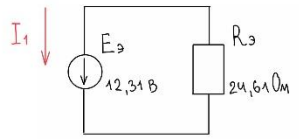

$$E_7 = J_7 R_{34567} = 0{,}57 \cdot 8{,}61 \approx 4{,}91 \text{ B}$$

6)




$$E_8 = E_2 + E_7 = 8 + 4{,}91 = 12{,}91 \text{ B}$$

$$E_9 = J_1 R_{34567} = 2 \cdot 8{,}61 = 17{,}22 \text{B}$$

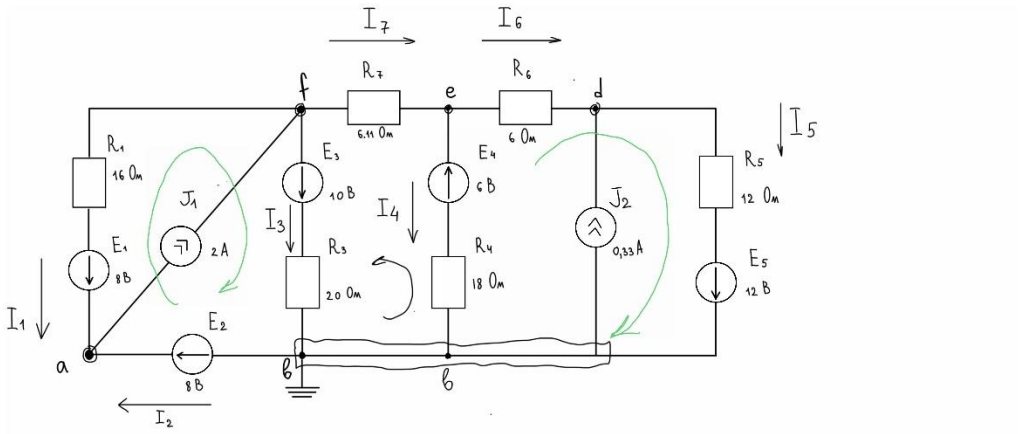



$$R_э = R_1 + R_{34567} = 16 + 8{,}61 = 24{,}61 \text{ Ом}$$

$$E_э = E_9 + E_1 - E_8 = 17{,}22 + 8 - 12{,}91 = 12{,}31 \text{ B}$$

$$I_1 = \frac{E_э}{R_э} = \frac{12{,}31}{24{,}61} \approx 0{,}5 \text{ A}$$

2. Рассчитать токи ветвей по законам Кирхгофа.




n (кол-во узлов) = 5

m (кол-во ветвей) = 7 - без ветвей с ист. тока

n − 1 = 4 уравнение по I з. К.

$$\begin{cases} I_1 + I_2 = J_1 \\ J_1 = I_1 + I_3 + I_7 \\ I_7 = I_4 + I_6 \\ I_6 + J_2 = I_5 \end{cases} \qquad \begin{cases} I_1 + I_2 = 2 \\ I_1 + I_3 + I_7 = 2 \\ -I_4 - I_6 + I_7 = 0 \\ I_5 - I_6 = 0{,}33 \end{cases}$$

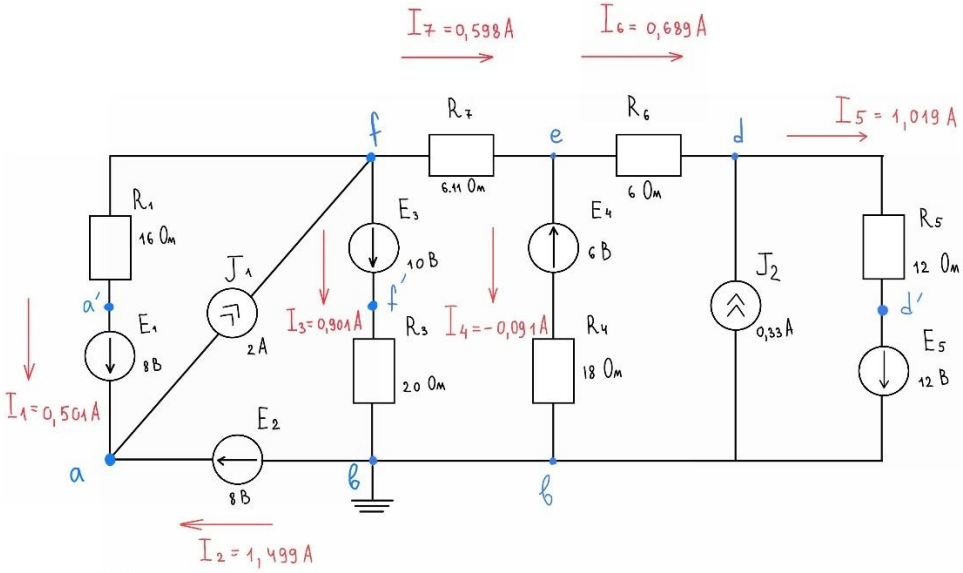
m − (n − 1) = 3 уравнение по II з. К.

$$\begin{cases} -E_1 + E_2 + E_3 = -I_1 R_1 + I_3 R_3 \\ E_4 + E_3 = -I_4 R_4 + I_3 R_3 - I_7 R_7 \\ E_4 + E_5 = -I_4 R_4 + I_6 R_6 + I_5 R_5 \end{cases} \qquad \begin{cases} -16 I_1 + 20 I_3 = 10 \\ 20 I_3 - 18 I_4 - 6.11 I_7 = 16 \\ -18 I_4 + 12 I_5 + 6 R_6 = 18 \end{cases}$$

$$\begin{array}{c} \begin{array}{ccccccc} I_1 & I_2 & I_3 & I_4 & I_5 & I_6 & I_7 \end{array} \\ \begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -1 & 0 & -1 & 1 \\ 0 & 0 & 0 & 0 & 1 & -1 & 0 \\ -16 & 0 & 20 & 0 & 0 & 0 & 0 \\ 0 & 0 & 20 & -18 & 0 & 0 & -6{,}11 \\ 0 & 0 & 0 & -18 & 12 & 6 & 0 \end{pmatrix} \end{array} \begin{pmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \\ I_5 \\ I_6 \\ I_7 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 0 \\ 0{,}33 \\ 10 \\ 16 \\ 18 \end{pmatrix}$$

$$\begin{pmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \\ I_5 \\ I_6 \\ I_7 \end{pmatrix} = \begin{pmatrix} 0{,}501 \\ 1{,}499 \\ 0{,}901 \\ -0{,}091 \\ 1{,}019 \\ 0{,}689 \\ 0{,}598 \end{pmatrix}$$

3. Построить потенциальную диаграмму для одного из контуров




$\varphi_б = 0$ В

$\varphi_a = \varphi_б + E_2 = 8$ В

$\varphi_{a'} = \varphi_a - E_1 = 8 - 8 = 0$ В

$\varphi_f = \varphi_{a'} + I_1 R_1 = 0 + 0{,}501 \cdot 16 \approx 8$ В

$\varphi_{f'} = \varphi_f + E_3 = 8 + 10$ В $= 18$ В

$\varphi_в = \varphi_{f'} - I_3 R_3 = 18 - 0{,}901 \cdot 20 = -0{,}002 \approx 0$ В

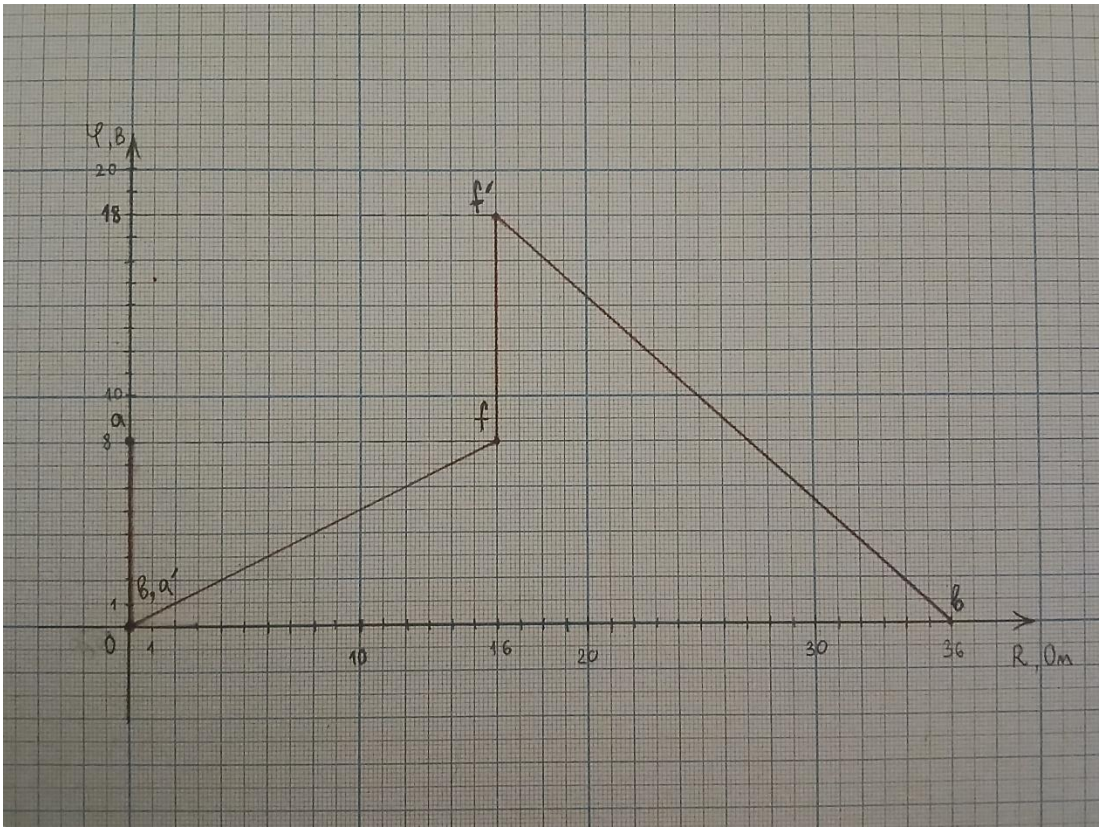
Для задачи 4:

$\varphi_e = \varphi_f - I_7 R_7 = 8 - 0{,}598 \cdot 6{,}11 \approx 4{,}3$ В

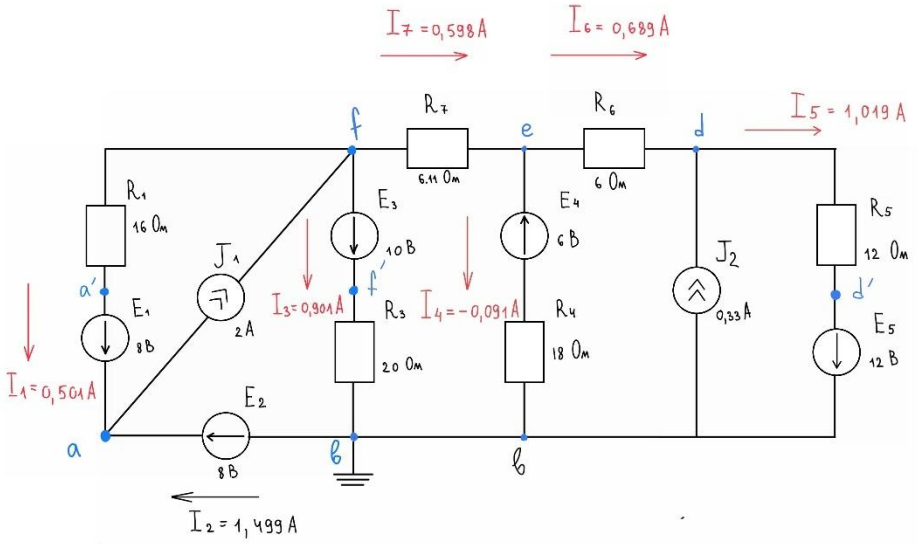
$\varphi_d = \varphi_e - I_6 R_6 = 4{,}3 - 0{,}689 \cdot 6 \approx 0{,}17$ В

$\varphi_{d'} = \varphi_d - I_5 R_5 = 0{,}17 - 1{,}019 \cdot 12 \approx -12$ В

$\varphi_в = \varphi_c = \varphi_{d'} + E_5 = -12 + 12 = 0$



4. Рассчитать баланс мощностей




$$\sum EI = E_1 I_1 + E_2 I_2 + E_3 I_3 - E_4 I_4 + E_5 I_5 + J_1(\varphi_f - \varphi_a) + J_2(\varphi_d - \varphi_c) =$$

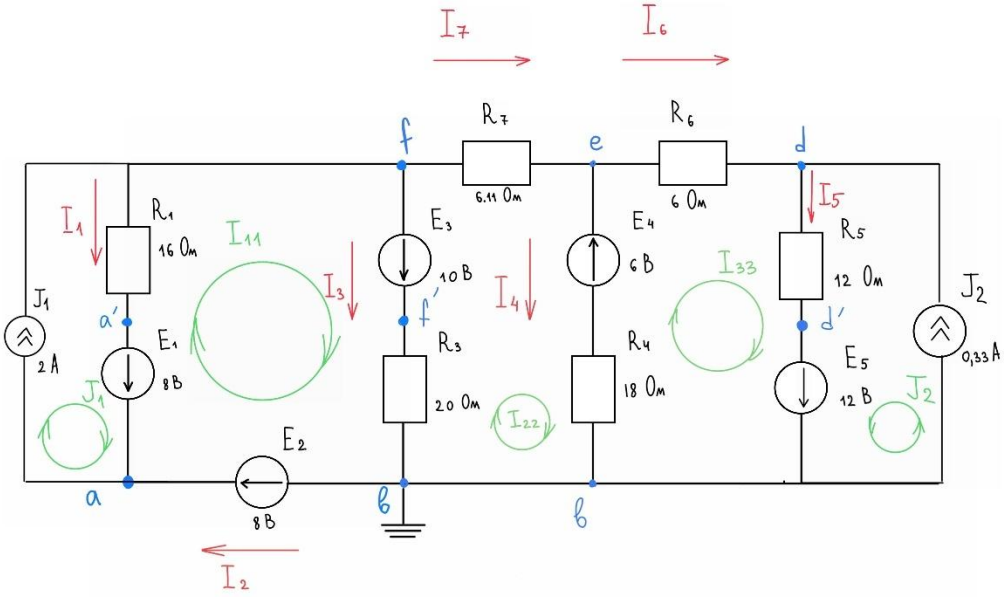
$$= 8 \cdot 0{,}501 + 8 \cdot 1{,}499 + 10 \cdot 0{,}901 + 6 \cdot 0{,}091 + 12 \cdot 1{,}019 + 2(8-8) + 0{,}33(0{,}17 - 0) = 37{,}8401 \text{ Вт}$$

$$\sum I^2 R = I_1^2 R_1 + I_3^2 R_3 + I_4^2 R_4 + I_5^2 R_5 + I_6^2 R_6 + I_7^2 R_7 =$$

$$= 0{,}501^2 \cdot 16 + 0{,}901^2 \cdot 20 + 0{,}091^2 \cdot 18 + 1{,}019^2 \cdot 12 + 0{,}689^2 \cdot 6 + 0{,}598^2 \cdot 6{,}11 = 37{,}89471 \text{ Вт}$$

$$37{,}8401 \text{ Вт} \sim 37{,}89471 \text{ Вт}$$

5. Рассчитать токи ветвей методом контурных токов




Кол-во уравнений: $m-n+1 = 7-5+1 = 3$

$$\begin{cases} I_{11}(R_1+R_3) - I_{22}R_3 - J_1R_1 = E_3 + E_2 - E_1 \\ I_{22}(R_3+R_4+R_7) - I_{11}R_3 - I_{33}R_4 = -E_3 - E_4 \\ I_{33}(R_4+R_5+R_6) - I_{22}R_4 + J_2R_5 = E_4 + E_5 \end{cases}$$

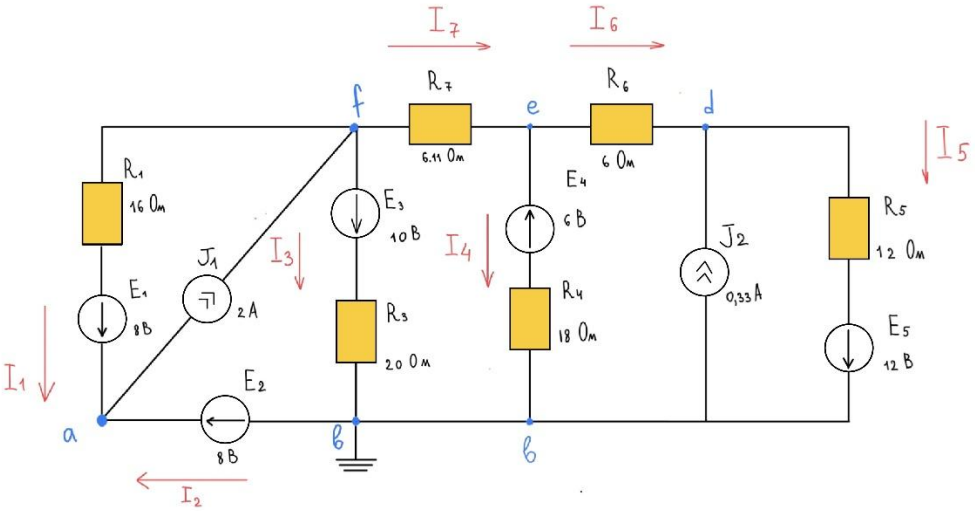
$$\begin{cases} 36I_{11} - 20I_{22} - 2\cdot 16 = 10 \\ 44{,}11I_{22} - 20I_{11} - 18I_{33} = -16 \\ 36I_{33} - 18I_{22} + 0{,}33\cdot 12 = 18 \end{cases}$$

$$\begin{cases} 36I_{11} - 20I_{22} = 42 \\ 44{,}11I_{22} - 20I_{11} - 18I_{33} = -16 \\ 36I_{33} - 18I_{22} = 14{,}04 \end{cases}$$

$$\begin{cases} I_{11} = 1{,}499 \\ I_{22} = 0{,}598 \\ I_{33} = 0{,}689 \end{cases}$$

$$\begin{cases} I_1 = J_1 - I_{11} = 2 - 1{,}499 = 0{,}501 \\ I_2 = I_{11} = 1{,}499 \\ I_3 = I_{11} - I_{22} = 1{,}499 - 0{,}598 = 0{,}901 \\ I_4 = I_{22} - I_{33} = 0{,}598 - 0{,}689 = -0{,}091 \\ I_5 = I_{33} + J_2 = 0{,}689 + 0{,}33 = 1{,}019 \\ I_6 = I_{33} = 0{,}689 \\ I_7 = I_{22} = 0{,}598 \end{cases}$$

6. Рассчитать токи ветвей методом узловых потенциалов




Пусть $\varphi(в) = 0$, тогда $\varphi(a) = E_2 = 8$ В

$$gf = \frac{1}{R_1} + \frac{1}{R_3} + \frac{1}{R_7} = \frac{1}{16} + \frac{1}{20} + \frac{1}{6{,}11} = \frac{13499}{48880} \text{ См}$$

$$ge = \frac{1}{R_7} + \frac{1}{R_6} + \frac{1}{R_4} = \frac{1}{6{,}11} + \frac{1}{6} + \frac{1}{18} = \frac{2122}{5499} \text{ См}$$

$$gd = \frac{1}{R_6} + \frac{1}{R_5} = \frac{1}{6} + \frac{1}{12} = \frac{1}{4} \text{ См}$$

$$gfe = \frac{1}{R_7} = \frac{1}{6{,}11} \approx \frac{1}{6{,}11} \text{ См}$$

$$ged = \frac{1}{R_6} = \frac{1}{6} \text{ См}$$

$$gfd = 0$$

$$Jf = J1 - \frac{E_3}{R_3} = 2 - \frac{10}{20} = 1{,}5 \text{ А}$$

$$Je = \frac{E_4}{R_4} = \frac{6}{18} = \frac{1}{3} \text{ А}$$

$$Jd = J_2 - \frac{E_5}{R_5} = 0{,}33 - \frac{12}{12} = -0{,}67 \text{ А}$$

$$\begin{cases} \varphi(f)\cdot g(f) - \varphi(e)\, g(fe) - \varphi(d)\, g(fd) = Jf \\ \varphi(e)\, g(e) - \varphi(f)\, g(fe) - \varphi(d)\, g(ed) = Je \\ \varphi(d)\, g(d) - \varphi(f)\, g(fd) - \varphi(e)\, g(ed) = Jd \end{cases}$$

$$\begin{cases} \frac{13499}{48880}\cdot\varphi f - \frac{1}{6{,}11} e = 1{,}5 \\ \frac{2122}{5499}\cdot\varphi e - \frac{1}{6{,}11}\cdot\varphi f - \frac{1}{6}\varphi d = \frac{1}{3} \\ 0{,}250\cdot\varphi d - \frac{1}{6}\varphi e = -0{,}67 \end{cases}$$

$$\begin{pmatrix} \frac{13499}{48880} & -\frac{1}{6,11} & 0 \\ -\frac{1}{6,11} & \frac{2122}{5499} & -\frac{1}{6} \\ 0 & -\frac{1}{6} & \frac{1}{4} \end{pmatrix} \begin{pmatrix} \varphi f \\ \varphi e \\ \varphi d \end{pmatrix} = \begin{pmatrix} 1,5 \\ \frac{1}{3} \\ -0,67 \end{pmatrix}$$

$$\begin{cases} \varphi e = \frac{173160}{21599} \approx 8,017\ B \\ \varphi d = 4,363\ B \\ \varphi f = 0,228\ B \end{cases}$$

$$I_1 = \frac{\varphi f - \varphi a + E_1}{R_1} = \frac{8,017}{16} \approx 0,501\ A$$

$$I_3 = \frac{\varphi f - \varphi b + E_3}{R_3} = \frac{8,017+10}{20} \approx 0,901\ A$$

$$I_4 = \frac{\varphi e - \varphi b - E_4}{R_4} = \frac{4,363-6}{18} \approx -0,091\ A$$

$$I_5 = \frac{\varphi d - \varphi b + E_5}{R_5} = \frac{\varphi d + 12}{12} \approx \frac{0,228+12}{12} = 1,019 A$$

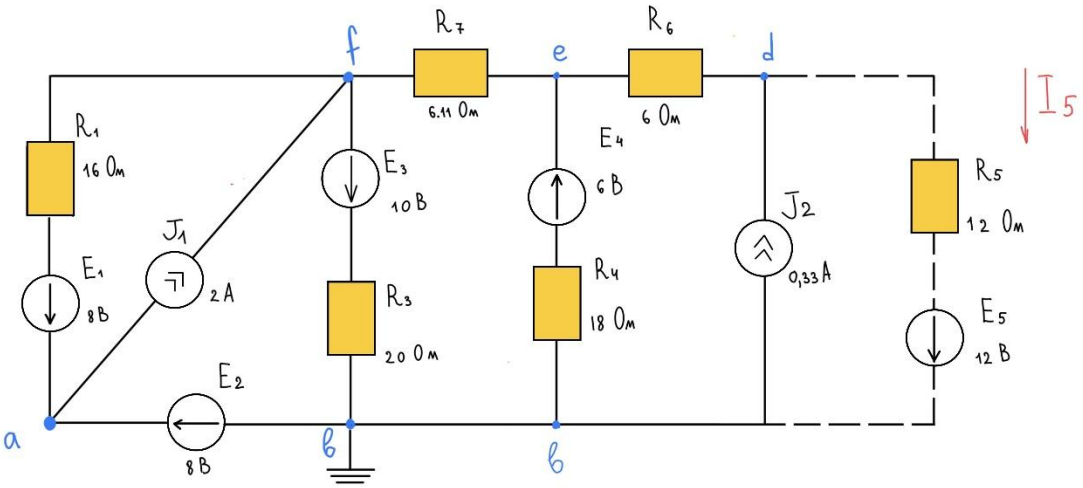
$$I_6 = \frac{\varphi e - \varphi d}{R_6} = \frac{4,363-0,228}{6} \approx 0,689\ A$$

$$I_7 = \frac{\varphi f - \varphi e}{R_7} = \frac{8,017-4,363}{6,11} \approx 0,598\ A$$

$I_1 + I_2 = J_2$ по I закону Кирхгофа

$I_2 = J_2 - I_1 = 2 - 0,501 = 1,499\ A$

7. Рассчитать ток в одной из ветвей методом эквивалентного генератора




Найдем ток $I_5$:

Пусть $\varphi(в) = 0$, тогда $\varphi(a) = E_2 = 8$ В

$$g_f = \frac{1}{R_1} + \frac{1}{R_3} + \frac{1}{R_7} = \frac{1}{16} + \frac{1}{20} + \frac{1}{6{,}11} = \frac{13499}{48880} \text{ См}$$

$$g_e = \frac{1}{R_7} + \frac{1}{R_6} + \frac{1}{R_4} = \frac{1}{6{,}11} + \frac{1}{6} + \frac{1}{18} = \frac{2122}{5499} \text{ См}$$

$$g_d = \frac{1}{R_6} = \frac{1}{6} \text{ См}$$

$$g_{fe} = \frac{1}{R_7} = \frac{1}{6{,}11} \approx \frac{1}{6{,}11} \text{ См}$$

$$g_{ed} = \frac{1}{R_6} = \frac{1}{6} \text{ См}$$

$$g_{fd} = 0$$

$$J_f = J_1 - \frac{E_3}{R_3} = 2 - \frac{10}{20} = 1{,}5 \text{ А}$$ ($E_1$ не берем т.к. $\varphi$ известен)

$$J_e = \frac{E_4}{R_4} = \frac{6}{18} = \frac{1}{3} \text{ А}$$

$$J_d = J_2 = 0{,}33 \text{ А}$$

$$\begin{cases} \varphi(f)\cdot g(f) - \varphi(e)\, g(fe) - \varphi(d)\, g(fd) = J_f \\ \varphi(e)\, g(e) - \varphi(f)\, g(fe) - \varphi(d)\, g(ed) = J_e \\ \varphi(d)\, g(d) - \varphi(f)\, g(fd) - \varphi(e)\, g(ed) = J_d \end{cases}$$

$$\begin{cases} \frac{13499}{48880} \cdot \varphi f - \frac{1}{6{,}11} e = 1{,}5 \\ \frac{2122}{5499} \cdot \varphi e - \frac{1}{6{,}11} \cdot \varphi f - \frac{1}{6} \varphi d = \frac{1}{3} \\ \frac{1}{6} \cdot \varphi d - \frac{1}{6} \varphi e = 0{,}3 \end{cases}$$

$$\begin{pmatrix} \frac{13499}{48880} & -\frac{1}{6{,}11} & 0 \\ -\frac{1}{6{,}11} & \frac{2122}{5499} & -\frac{1}{6} \\ 0 & -\frac{1}{6} & \frac{1}{6} \end{pmatrix} \begin{pmatrix} \varphi f \\ \varphi e \\ \varphi d \end{pmatrix} = \begin{pmatrix} 1{,}5 \\ \frac{1}{3} \\ 0{,}33 \end{pmatrix}$$

$$\begin{cases} \varphi e = \frac{38484}{29699} \\ \varphi d = 12{,}7 \\ \varphi f = 14{,}68 \end{cases}$$

$E_{эг} = \varphi d - \varphi c = 14{,}68 - 0 = 14{,}68 В$

Найдем $R_{эг}$:

$R_{13} = \frac{R_1 R_3}{R_1 + R_3} = \frac{16 \cdot 20}{16 + 20} = \frac{80}{9}$ Ом

$R_{137} = R_{13} + R_7 = \frac{80}{9} + 6{,}11 = 14{,}99889$ Ом

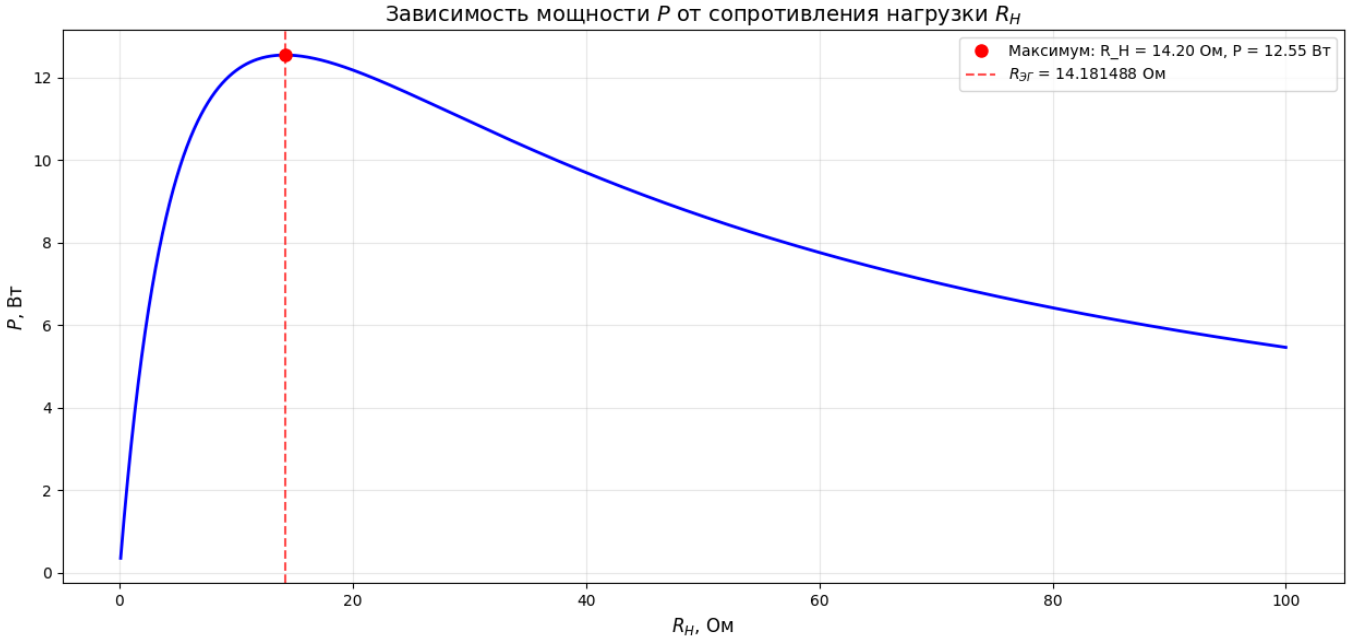
$R_{1347} = \frac{R_{137} \cdot R_4}{R_{137} + R_4} = \frac{14{,}99889 \cdot 18}{14{,}99889 + 18} = 8{,}181488$ Ом

$R_{13467} = R_6 + R_{1347} = 8{,}181488 + 6 = 14{,}181488$ Ом

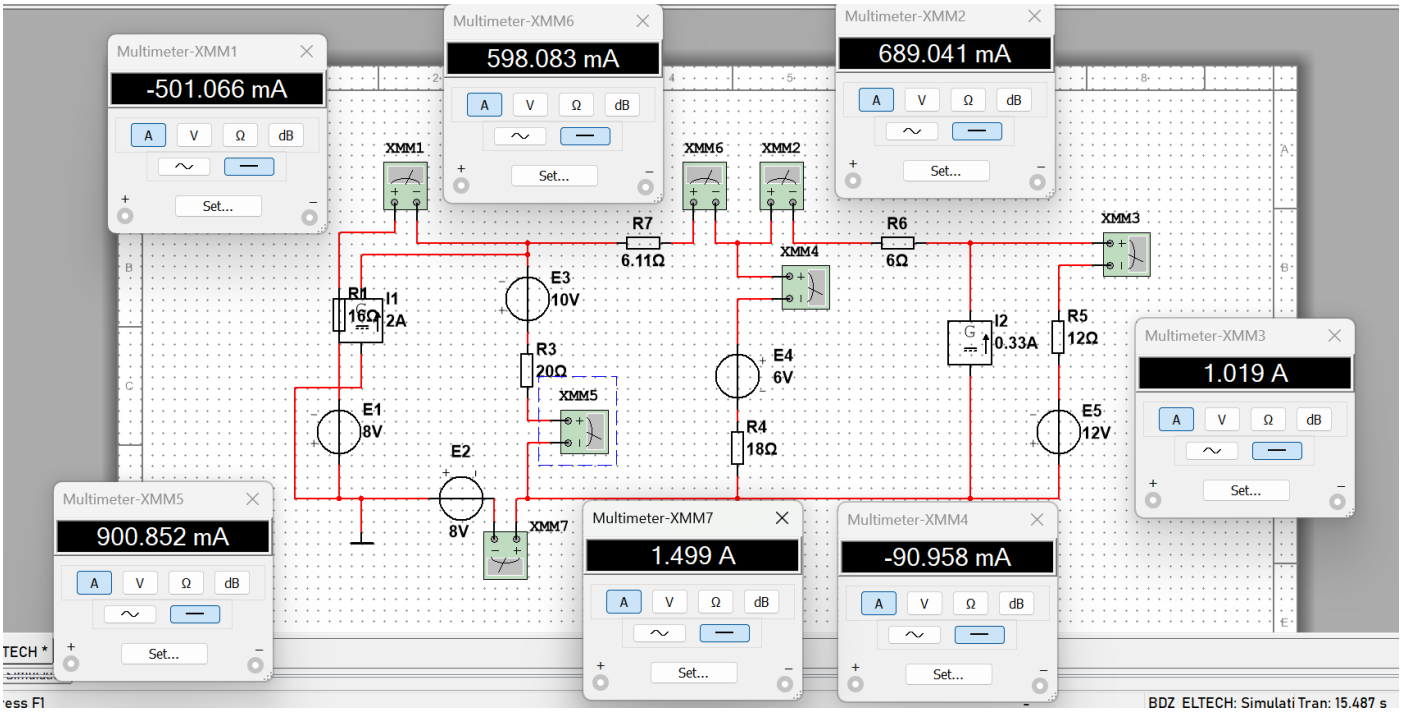
$R_{эг} = R_{13467} = 14{,}181488$ Ом

$I_5 = \frac{E_{эг} + E_5}{R_{эг} + R_5} = \frac{14{,}68 + 12}{14{,}181488 + 12} \approx 1{,}019$ A

Меняя сопротивление выбранной ветви, построить график зависимости мощности от сопротивления нагрузки




8. Собрать схему в программе Multisim и измерить токи



9. Составить таблицу с расчетными данными

| | МЭП | МНПЗК | МКТ | МУП | МЭГ | Multisim |
|---|---|---|---|---|---|---|
| $I1$ | 0.5 | 0.501 | 0.501 | 0.501 | | 0.501066 |
| $I2$ | | 1.499 | 1.499 | 1.499 | | 1.499 |
| $I3$ | | 0.901 | 0.901 | 0.901 | | 0.900852 |
| $I4$ | | 0.091 | 0.091 | 0.091 | | 0.090958 |
| $I5$ | | 1.019 | 1.019 | 1.019 | 1.019 | 1.019 |
| $I6$ | | 0.689 | 0.689 | 0.689 | | 0.689041 |
| $I7$ | | 0.598 | 0.598 | 0.598 | | 0.598083 |